

## Section 6.3 Exercises

Directions: For exercises 1 - 9, evaluate the expression.

1.  $\sin^{-1}\left(\frac{\sqrt{2}}{2}\right)$

2.  $\sin^{-1}\left(-\frac{1}{2}\right)$

3.  $\cos^{-1}\left(\frac{1}{2}\right)$

4.  $\cos^{-1}\left(-\frac{\sqrt{2}}{2}\right)$

5.  $\tan^{-1}(1)$

6.  $\tan^{-1}(-\sqrt{3})$

7.  $\tan^{-1}(-1)$

8.  $\tan^{-1}(\sqrt{3})$

9.  $\tan^{-1}\left(\frac{-1}{\sqrt{3}}\right)$

Directions: For exercises 10 - 14, use a calculator to evaluate each expression. Express all answers in radians rounded to two decimal places.

10.  $\cos^{-1}(-0.4)$

11.  $\arcsin(0.23)$

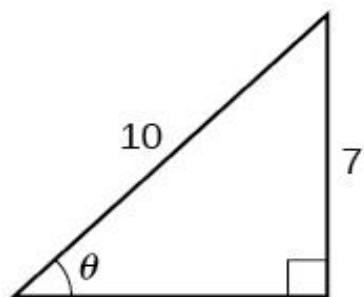
12.  $\arccos\left(\frac{3}{5}\right)$

13.  $\cos^{-1}(0.8)$

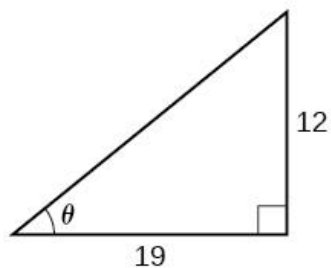
14.  $\tan^{-1}(6)$

Directions: For exercises 15 - 16, find the angle  $\theta$  in the given right triangle. Express all answers in degrees rounded to one decimal place.

15.



16.



Directions: For exercises 17 - 25, find the exact value, if possible, without a calculator. If it is not possible, explain why and state your answer in radians rounded to two decimal places.

17.  $\sin^{-1}(\cos(\pi))$

18.  $\tan^{-1}(\sin(\pi))$

19.  $\cos^{-1}\left(\sin\left(\frac{\pi}{3}\right)\right)$

20.  $\tan^{-1}\left(\sin\left(\frac{\pi}{3}\right)\right)$

21.  $\sin^{-1}\left(\cos\left(\frac{-\pi}{2}\right)\right)$

22.  $\tan^{-1}\left(\sin\left(\frac{4\pi}{3}\right)\right)$

23.  $\sin^{-1}\left(\sin\left(\frac{5\pi}{6}\right)\right)$

24.  $\tan^{-1}\left(\sin\left(\frac{-5\pi}{2}\right)\right)$

25.  $\cos\left(\sin^{-1}\left(\frac{1}{2}\right)\right)$

Directions: For exercises 26 -29, find the exact value without a calculator.

26.  $\cos\left(\sin^{-1}\left(\frac{4}{5}\right)\right)$

27.  $\sin\left(\cos^{-1}\left(\frac{3}{5}\right)\right)$

28.  $\sin\left(\tan^{-1}\left(\frac{4}{3}\right)\right)$

29.  $\cos\left(\tan^{-1}\left(\frac{12}{5}\right)\right)$

Directions: For exercises 30 - 33, find the exact value of the expression in terms of  $x$ .

30.  $\tan(\sin^{-1}(x - 1))$

31.  $\sin(\cos^{-1}(1 - x))$

32.  $\cos(\sin^{-1}(\frac{1}{x}))$

33.  $\cos(\tan^{-1}(3x - 1))$

34. Evaluate the following expression without using a calculator. State the exact value.

$$\frac{\sin^{-1}\left(\frac{1}{2}\right) - \cos^{-1}\left(\frac{\sqrt{2}}{2}\right) + \sin^{-1}\left(\frac{\sqrt{3}}{2}\right) - \cos^{-1}(1)}{\cos^{-1}\left(\frac{\sqrt{3}}{2}\right) - \sin^{-1}\left(\frac{\sqrt{2}}{2}\right) + \cos^{-1}\left(\frac{1}{2}\right) - \sin^{-1}(0)}$$

Directions: For exercises 35 - 39, find the exact value of the expression if  $\sin t = \frac{x}{x+1}$ .

35.  $\cos t$

36.  $\sec t$

37.  $\cot t$

38.  $\cos(\sin^{-1}(\frac{x}{x+1}))$

39.  $\tan^{-1}(\frac{x}{\sqrt{2x+1}})$